

Δ -Learning of High-Fidelity Electronic Structure Using Graph Neural Networks with Modified Node-Level Features

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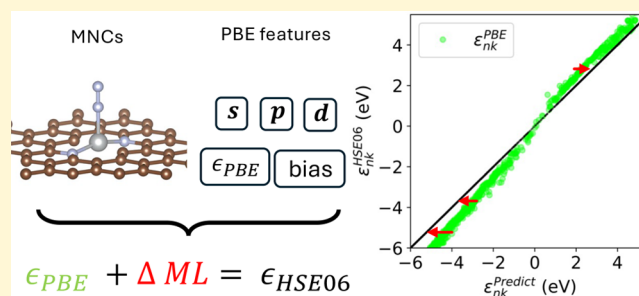


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ABSTRACT: In this work, we present a Δ -learning approach for predicting the eigenvalues calculated with the hybrid functional HSE06 (ϵ_{nk}^{HSE}) for a set of metal and nitrogen doped graphene catalysts (MNCs) from Perdew-Burke-Ernzerhof (PBE) inputs. The model presented here incorporates electronic scalar features along with structural information in a graph neural network (GNN). In particular, the PBE eigenvalues for different bands and k -points and orbital-resolved projectors are combined with the applied potential as node-level features along with structural information within the Atomistic Line Graph Neural Network (ALIGNN) architecture. These features enable flexibility for systems with electrified interfaces, such as in electrocatalysts and achieves mean absolute error (MAE) of less than 0.1 eV. The machine learning model reported here achieves a strong generalization to left-out adsorbates (MAE = 0.074 eV) and leave-one-chemical-space-out (MAE = 0.08 eV) and completely left-out metals (MAE = 0.072 eV), confirming the robustness of the machine learning (ML) model in predicting ϵ_{nk}^{HSE} .



Electrocatalysts are crucial for a variety of industries, especially those related to batteries, fuel cells, and water electrolysis. There is a high demand for electrocatalysts that can operate effectively at low overpotentials and high faradaic efficiencies. In recent years, materials consisting of graphene doped with dispersed metal–nitrogen sites (MNCs) have shown promise as catalysts for various electrocatalytic processes due to their tunability, high atom-utilization, and relatively cheap cost of manufacturing.^{1,2} MNCs are typically modeled as a single metal atom coordinated to 4 pyridinic nitrogen atoms embedded in graphene. However, recent studies have shown that the catalytic performance of such sites can be modulated by substituting C atoms for the N atoms to produce MN_x ($x = 1, 2, 3, 4$) sites.^{3,4} The metal center itself can also be optimized for a target application. Co/N-doped graphene has shown excellent results for electrocatalytic hydrogen evolution;⁵ Fe/N- and Ni/N-doped graphenes are promising for the reduction of CO₂ to CO with high faradaic efficiencies,^{6,7} and Fe/N-doped graphene is considered the top alternative to platinum for the oxygen reduction reaction in fuel cells.⁸

Machine learning (ML) has been used to predict the properties of catalysts directly⁹ as well as combined with quantum mechanical (QM) methods, such as density functional theory (DFT), to accelerate catalyst screening by enabling rapid evaluation of active sites and revealing structure–activity relationships. For example, Hammer and Norskov¹⁰ identified the relation between the DFT-calculated adsorption energies and the d -band energy of the electronic density of states (DOS). Although the d -band model has been successful in predicting reactivities and catalysts, it is limited to specific catalysts (e.g., metallic cases with surface d -states).

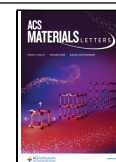
Despite the recent successful history of DFT, there still exist issues regarding its accuracy in describing the interaction of

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adsorbates on transition metal surfaces.^{11–15} This is in part due to inherent weakness of semilocal functionals (e.g., GGA) that have a noticeable self-interaction error (SIE),¹⁶ which leads to errors in the prediction of band gaps. This can subsequently influence the hybridization of the electronic orbitals on the surface and adsorbates and results in incorrect predictions for adsorption energies,¹⁷ motivating the need for higher-fidelity methods for calculating the relevant properties for screening catalysts.

One of the common strategies to reduce SIE in semilocal DFT is to use hybrid functionals instead that include exact exchange. Among them, the short-range hybrid Heyd–Scuseria–Ernzerhof (HSE06) functional has been shown to perform well for periodic systems, but its application to large, extended systems comes at a computational cost substantially higher than that of GGA functionals, which is impractical in many cases.

An alternative approach involves using ML to estimate properties calculated by the HSE06 hybrid functional, thereby overcoming computational bottlenecks. In previous work, we applied Δ -learning to accurately predict the HSE06 eigenvalues of bulk materials.¹⁸ As shown previously for Δ -learning of ϵ_{nk}^{HSE} ,¹⁸ the projection of each atomic orbital at every k -point is a key feature. However, in that work, the density of states (DOS) projected onto spatial orbitals was not atom-specific but averaged over all atoms for each material. Moreover, the model did not incorporate structural information. This approximation still yielded accurate results because of the largely homogeneous set of structures studied in that work.

Instead, a particularly effective way to represent material structures is through graph-based models, where atomic sites serve as nodes and edges represent “bonds” to neighbors within a cutoff distance between sites. Graph-based representations are well-suited for capturing structure-dependent properties, as they inherently encode atomic positions and interatomic distances, making them ideal inputs for neural networks. This approach was first demonstrated for materials science applications with the Crystal Graph Convolutional Neural Network (CGCNN), which efficiently learns structural features directly from the atomic positions.¹⁹ Message Passing Neural Networks (MPNNs)²⁰ have particularly improved the accuracy of graph-based representations for materials. Numerous MPNN variants have been published, each addressing specific challenges for property predictions and material design.^{21–33} In terms of modeling heterogeneous catalysts specifically, several MPNNs have been applied to the Open Catalysis Project data set³⁴ such as GemNetdT,²² DimeNet+,²⁷ PaiNN,²³ GemNet-OC,³⁵ SCN,³⁶ and EquiformerV2.³⁷

While MPNNs are designed to encode structural information, they do not intrinsically capture the electronic properties. Given the importance of electronic structure in heterogeneous catalysis (e.g., the d -band center model), Fung et al.³⁸ proposed an alternative approach using a Convolutional Neural Network (CNN) entitled DOSnet. Instead of relying on explicit structural representations, DOSnet extracts features directly from the density of states (DOS) to predict adsorption energies, offering a complementary strategy for modeling catalytic behavior. In other work, Wang et al. used a theory-infused neural network (TinNet), combined with d -band theory of chemisorption, to predict the reactivity of transition metal surfaces.³⁹

In this work, we present an MPNN framework that incorporates both structural and electronic properties as inputs

into a single ML model to enable the Δ -learning of HSE06 eigenvalues (ϵ_{nk}^{HSE}) to predict the entire band structure. Of particular importance for their application as electrocatalysts, our model also uses the bias voltage as a global property. In addition to the projectors, because the electronic structure of each GC-DFT calculation in the training set can be nontrivially impacted by the applied bias beyond a rigid shift of states above or below the Fermi level, we included it as a node-level feature as well. For the purpose of testing this methodology, we applied it to MNCs only. However, given that the ML model uses a graph-based representation and PBE-level features that are broadly transferable, it should generalize to other material classes (e.g., metal-oxide catalysts).

Our analysis shows that the proposed graph neural network (GNN) achieves promising results in the prediction of ϵ_{nk}^{HSE} for several generalization tasks, such as unseen adsorbates (MAE = 0.074 eV), unseen metals (MAE = 0.072 eV), and unseen MNCs (MAE = 0.09 eV). These results prove the model's ability to Δ -learn ϵ_{nk}^{HSE} much more accurately than the MAE between ground truth ϵ_{nk}^{HSE} and ϵ_{nk}^{PBE} , which is 0.91 eV. With this accuracy, this model can be used to predict HSE06-quality electronic structures and combined with other calculation postprocessing tools, such as projected crystal orbital Hamilton population (pCOHP) bonding analysis,⁴⁰ to study adsorbate bonding in MNC electrocatalysts.

For the construction of the data set, grand-canonical DFT (GC-DFT) calculations were performed on the MNC structures with a single metal-atom coordinated to 1, 2, 3, or 4 nitrogen atoms (indicated by MN₁C–MN₄C) embedded into a two-dimensional graphene monolayer. As shown in Table 1, the data set consists of 6 metal centers (Co, Cu, Fe,

Table 1. Summary of All Structure/Calculation Parameters Varied to Create the Training Dataset^a

Structure Category	Target Data Set
Active sites	MN ₁ C, MN ₂ C, MN ₃ C, MN ₄ C
Metal centers	Co, Cu, Fe, Mo, Ni, Pt
Bias voltages	0.0 V, −0.5 V, −1.0 V
Adsorbates	CO, CO ₂ , COOH, H, N ₂ H, NH ₃ , N ₂

^aThe different combinations of active sites, metal centers, adsorbates, and bias voltages resulted in 168 unique structures with 504 unique calculations (3 bias voltages per structure).

Mo, Ni, Pt), 7 adsorbates (CO, CO₂, COOH, H, N₂, N₂H, and NH₃), and 3 bias voltages (0.0, −0.5, and −1.0 V vs SHE), which results in a total of 168 unique MNC structures and 504 unique calculations. The 7 selected adsorbates encompass important intermediates to the widely studied electrochemical hydrogen evolution, CO₂ reduction, and nitrogen reduction reactions. For every calculation, we collect eigenvalues at every k -point and every band n (ϵ_{nk}^{PBE}), their projections P_{nk}^i on every atom i and spatial orbital s , and their corresponding ϵ_{nk}^{HSE} from HSE06 calculations. The projection scheme used here to generate features relating to each spatial orbital at every k -point is summarized in our previous publication,¹⁸ with other details about the DFT are provided in the Computational Methods.

All ϵ_{nk}^{PBE} and ϵ_{nk}^{HSE} values were shifted by the Fermi energy such that the Fermi level is at 0 eV. Only ϵ_{nk}^{PBE} within ± 10 eV of the Fermi level were included in the model training because eigenvalues beyond this range have smaller contributions to catalytic activity and would disproportionately affect the loss function when tuning the ML model.

Unlike typical applications of GNNs in materials, however, where each sample is used to generate a different graph that is in turn mapped to a unique target property, our goal here is to predict ϵ_{nk}^{HSE} for a given ϵ_{nk}^{PBE} and P_{nk}^{is} . Thus, each structure in our data set has multiple target properties. Our data set is composed of only 168 unique structures (504 unique calculations), whereas the number of target properties (i.e., ϵ_{nk}^{HSE}) is on the order of 10^5 .

The GNN framework for Δ -learning ϵ_{nk}^{HSE} requires modifying the Atomistic Line Graph Neural Network (ALIGNN)²⁴ architecture to incorporate the atomic projectors as node-level features, in addition to the local bonding information it natively captures. These node-level features replace the default values used in ALIGNN, which are based on scalar atomic properties from the Matminer feature set.²⁴ It should be noted that the conventional ALIGNN architecture model that is trained using typical node-level features obtained from Matminer cannot distinguish between eigenvalues at different k -points and predicts all the eigenvalues to be the same (see the parity plot of $\epsilon_{nk}^{Predict}$ and ϵ_{nk}^{HSE} in Figure S2). In addition to the projectors, because the electronic structure of each GC-DFT calculation in the training set can be nontrivially impacted by the applied bias beyond a rigid shift of states above or below the Fermi level, we have included it as a node-level feature as well.

The key features affecting model accuracy were identified through a feature ablation analysis (Table 2). Due to the

Table 2. MAE of the HSE06 Eigenvalues, ϵ_{nk}^{HSE} , for an Ablation Study Using Different Sets of Node Level Features in Model Training^a

Left-out Adsorbate(s)	Node Level Feature Set				
	P_{nk}^{is}	ϵ_{nk}^{PBE}	$P_{nk}^{is} + \text{Bias}$	$P_{nk}^{is} + \epsilon_{nk}^{PBE}$	$P_{nk}^{is} + \epsilon_{nk}^{PBE} + \text{Bias}$
CO	0.340	0.080	0.228	0.067	0.069
CO ₂	0.391	0.116	0.296	0.076	0.085
COOH	0.276	0.087	0.185	0.069	0.074
H	0.528	0.093	0.404	0.076	0.076
N ₂	0.340	0.093	0.232	0.086	0.074
N ₂ H	0.381	0.081	0.268	0.066	0.063
NH ₃	0.414	0.102	0.372	0.070	0.075
N adsorbates	0.506	0.087	0.397	0.080	0.073
C adsorbates	0.410	0.090	0.339	0.082	0.084

^aEvery row shows the MAE toward predicting ϵ_{nk}^{HSE} for each target adsorbate when those structures were excluded from the training set. Only MN₁C structures were included here. All values are in eV.

computational cost, this analysis was limited to only the MN₁C structures, but the trends observed here are expected to extend to the other MNC structures. Broadly, we find that including ϵ_{nk}^{PBE} as node-level features is the most important and that they substantially improve model performance. Additional improvement is gained from including the projectors as node-level features as well, however, the improvement is smaller. Overall, the best performing model includes the P_{nk}^{is} , ϵ_{nk}^{PBE} , and bias features as node-level features (Figure 1), with errors ranging from 0.06 to 0.09 eV. We also considered using ϵ_{nk}^{PBE} and/or bias as global features; however, their performances are poorer than the model using ϵ_{nk}^{PBE} , bias, and P_{nk}^{is} as node-level features. Further details on the modified ALIGNN architecture and global feature analysis are provided in SI Sections I and II, respectively. The modified ALIGNN code is provided in the

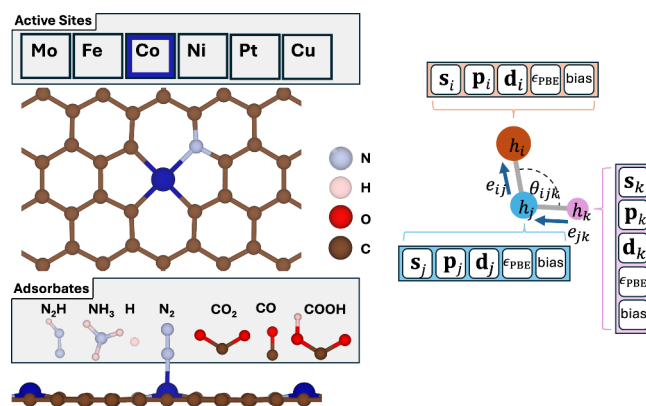


Figure 1. Schematic of the architecture and data set used in this work. A typical MN₁C surface with Co is depicted with the set of metals and adsorbates studied in this work. The crystal graph and line graph consist of the node features h_i including the s , p and d P_{nk}^{is} , ϵ_{nk}^{PBE} , and bias voltage, the edge features e_{ij} and the angle θ_{ijk} . They are passed through 4 ALIGNN layers and 4 edge-gated GNN layers where l and t are numbers of layers.

github repository https://github.com/Sutton-Research-Lab/delta_learning_eigen/.

Another key factor influencing the accuracy of the ML model is the number of training samples. To establish a consistent training procedure, we systematically explored various training set sizes by considering 1%, 5%, 10%, and 25% of eigenvalues with their associated projectors selected per k -point, and the total number of irreducible k -points from which to include eigenvalue data (5 total symmetry-irreducible k -points could be chosen as the maximum). MAEs were averaged over three experiments with different randomly selected sets of eigenvalues.

For each test set, we predicted ϵ_{nk}^{HSE} for all systems with Fe as the metal center in an MN₁C site configuration. Figure 2

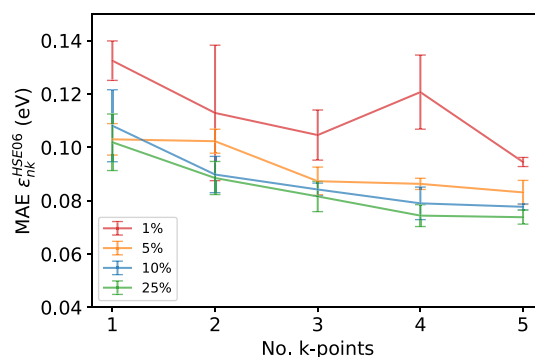


Figure 2. Learning curve for a test set that included all Fe samples from MN₁C systems as a function of percentage of eigenvalues between -10 and 10 eV, taken from different numbers of symmetry-irreducible k -points. For all of these experiments, no structure with Fe as the metal center was included in any training set.

shows that, with just 1% of the eigenvalues/projectors from 1 k -point, the ML model has a reasonable accuracy (MAE = 0.13 eV). Adding 5% of the eigenvalues/projectors from every k -point decreases the MAE by 0.03–0.04 eV. With 10% of the eigenvalues/projectors, the error of the model can be further reduced by 0.01 eV, but using a higher percentage (e.g., 25%) has a negligible impact on model performance. In addition, the

error bars, which can be up to 0.04 eV with 1% of eigenvalues/projectors, become less than 0.01 eV for 5% and higher percentages of eigenvalues/projectors.

Training on data from 2 k -points lowers the MAE by 0.02 eV relative to training on the 1 k -point. However, additional sampling from higher numbers of k -points provides negligible improvements in the model accuracy. Our observation that increasing the number of k -points only marginally improves the model is consistent with our previous work.¹⁸ Based on these results, we selected 10% of the eigenvalues/projectors for 4 k -points for subsequent analysis in this work.

To apply the model/architecture to new catalytic systems, the trained ML model should perform accurately for MNCs (i.e., structures, adsorbates, and metal centers) outside of the training set. Here, we show how several independently trained models perform these generalization tasks. This is particularly important when one is interested in applying the model/architecture to new catalytic systems, e.g., with metallic sites or adsorbates different from those studied here.

First, we tested the impact of only including systems with one type of metal atom center in the test set while keeping the remaining metal centers in the training set, resulting in six independently trained models. As shown in Figure 3, models applied to unseen metal centers achieve MAEs of 0.06–0.10 eV, matching the range of the best ML model in Figure 2. This result is robust in three different training sets with MAEs that differ by <0.02 eV.

In addition to the unseen metal centers, we examined the performance of the ML model for unseen adsorbates by only

including systems with one type of adsorbate in the test set while keeping the remaining ones in the training set (leading to seven independently trained models). Figure 3 shows that the MAEs for each of these models evaluated on the left-out adsorbate structures range from 0.06 to 0.09 eV, highlighting that this framework can be applied accurately to systems with new adsorbates not currently in this data set. More generally, this result is further supported by experiments where adsorbates in the same chemical space, i.e., carbon-based adsorbates (CO, COOH, and CO₂) and nitrogen-based adsorbates (N₂, N₂H, and NH₃), were excluded from the training data so that two independent models could be evaluated using test data from the unseen group. These models predicted $\epsilon_{nk}^{\text{HSE}}$ for the unseen adsorbate group with MAEs of 0.07 to 0.08 eV, which is similar to the best ML model in Figure 2.

As a final test of the ML model's generalization to samples left out of the training set, we trained a single model on, e.g., only structures containing the MN₁C active site and predicted $\epsilon_{nk}^{\text{HSE}}$ for all remaining MN _{x} C ($x = 2 - 4$) structures. This was repeated so that all MN _{x} C active sites were in the training set. MAEs are within 0.07 to 0.10 eV for these four test sets (see Table 3), indicating that a model trained to a single type of

Table 3. Summary Test MAEs for Models Trained on Data from Structures with One Type of MN _{x} C Active Site^a

Training Set	MN ₁ C	MN ₂ C	MN ₃ C	MN ₄ C
MN ₁ C		0.076	0.078	0.095
MN ₂ C	0.084		0.079	0.090
MN ₃ C	0.086	0.077		0.073
MN ₄ C	0.100	0.104	0.092	

^aOne type of MN _{x} C active site is in every row and applied to test data for structures with the other three types of MN _{x} C active sites in every column. All values in eV.

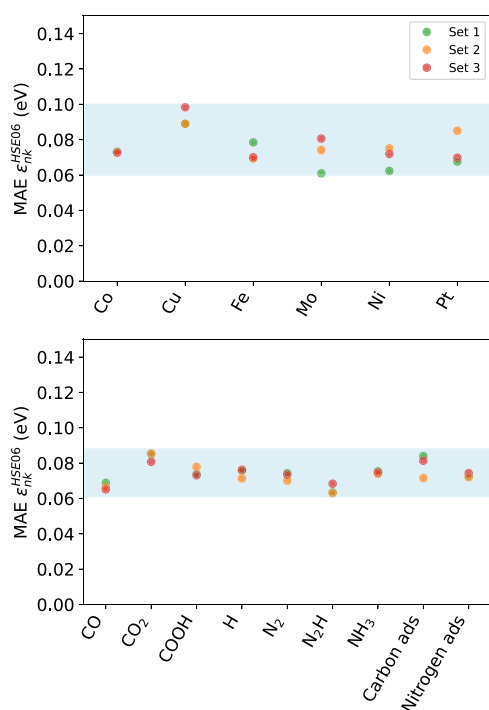


Figure 3. (Top) Leave-one metal out and **(Bottom)** leave-one adsorbate out experiments performed three times with three separate training sets selected using 10% of the PBE eigenvalues/projectors for 4 k -points in the MN₁C systems (indicated by different colors). The last two points on the x -axis correspond to leave-one-adsorbate-out experiments, where all nitrogen-based (N₂, N₂H, NH₃) or all carbon-based (CO, CO₂, and COOH) adsorbates were excluded from the training set.

MNC structure can be applied to other MNCs without a substantial loss in accuracy. This finding is particularly relevant as a variety of active site MNC structures have been proposed.^{41,42}

The ML-predicted eigenvalues can be used to postprocess calculations and to gain insight into how a hybrid functional will impact the system's electronic structure as compared to either a GGA and hybrid functional. PBE is known to systematically overbind covalently bonded adsorbates and underbind adsorbates bound via dispersion forces;⁴³ therefore, we expect shifts in the predicted HSE06 electronic structure relative to PBE for these samples. Thus, a key question is how well the ML model predicted the electronic structure compared with HSE06.

To test this, we used the ML model trained to the MN₁C systems (excluding Fe as the metal center) to predict the HSE06 eigenvalues for H, CO, NO, NH₂, O, and O₂ adsorbed on the Fe site in an FeN₄C catalyst with an applied bias of 0 V vs SHE. Of these six, NO, NH₂, O, and O₂ were absent from the training data set of 504 calculations described above. These systems were chosen due to the interest in the FeN₄C catalyst⁴⁴ and importance of these adsorbates to reactions such as the hydrogen evolution, CO₂ reduction, ammonia synthesis, and oxygen reduction reactions.

Overall, a great agreement between the HSE06 DOS and ML-predicted DOS is observed (e.g., an example of adsorption of CO on different metal centers is provided in SI Figure S3).

Moreover, ϵ_{nk}^{PBE} , ML-predicted eigenvalues, and ϵ_{nk}^{HSE} were combined with PBE projectors to perform pCOHP bonding analysis to evaluate how well the ML-predicted eigenvalues capture changes in (anti)bonding interactions.

pCOHP bonding analysis has been used to gain insight into a wide variety of materials and catalyzed reactions.⁴⁰ Thus, in this work, we use pCOHP as a proof of principle, but we expect these results to extend to similar approaches that extract bonding information from the electronic structure, such as COOP (crystal orbital overlap population) and COBI (crystal orbital bond index).

The predictions from the ML model compare well to the HSE06 pCOHP trends for all six adsorbates. For example, the pCOHP distributions for H and O₂ exhibit a shift of the occupied bonding and antibonding orbitals to Fe to lower energies with a smaller shift in the unoccupied state orbitals (see Figure 4). Figure S4 shows the results for all six

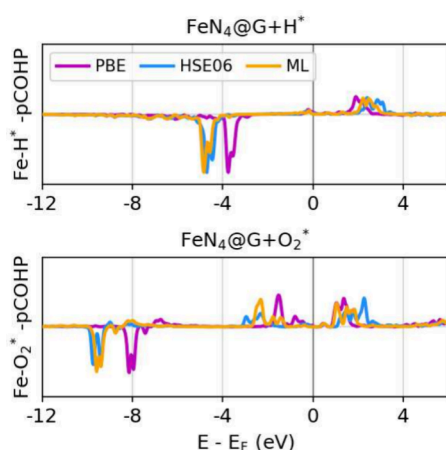


Figure 4. pCOHP distributions for H and O₂ adsorbed on the Fe site of the FeN₄ active site. The plots compare pCOHP spectra calculated using PBE eigenvalues and projectors (magenta), HSE06 eigenvalues and projectors (blue), and ML eigenvalues from our model and PBE projectors (orange).

adsorbates. The good performance of the ML predictions for out-of-distribution adsorbates such as O₂ (as well as NO, NH₂, and O) further suggests that the ML model generalizes well. Overall, these results indicate that the ML DOS is sufficiently accurate to capture trends in the electronic structure across different adsorbates and catalysts, which is particularly valuable for rationalizing reactivity trends in these systems.

Overall, we successfully utilized the ALIGNN model to Δ -learn HSE06 eigenvalues across a range of catalytically relevant energies around the Fermi level by utilizing PBE eigenvalues, projectors, and bias voltage as node-level features. We determined an optimal number of eigenvalues and k -points that could produce the desired results (MAEs of 0.06–0.09 eV) through conducting several leave-one-adsorbate out experiments. The ablation study showed that ϵ_{nk}^{PBE} , P_{nk}^{is} , and bias voltage as node features without any global feature substantially outperform the default ALIGNN architecture that uses atomic properties as node-level features. This model generalizes well across held-out metal center atoms, held-out stoichiometries of MN_xC sites, and held-out adsorbates, including when grouped sets of carbon- or nitrogen-based adsorbates were only in the test set. Our detailed analysis of left-out samples shows that a single model can be applied to a

wide range of out-of-distribution MNC systems without a loss in accuracy and used in additional bonding analysis for catalytic systems in good agreement with HSE06. This framework is general and can be applied to other electrocatalysts of interest across a wide-range of applied biases, paving the way for future models that are aware of both atomic and electronic structure impacts to electrocatalyst performance.

COMPUTATIONAL METHODS

Ionic optimizations were performed using GC-DFT at bias voltages of 0.0, −0.5, and −1.0 V vs SHE in the JDFTx software package⁴⁵ with a plane-wave energy and charge-density cutoffs of 20 and 100 hartree, respectively, a k -point density of at least 1000 k -points per Å^{−1}, the CANDLE solvation model,^{46,47} the PBE functional with Grimme's D3 dispersion correction,^{48,49} and ultrasoft GBRV Vanderbilt pseudopotentials.⁵⁰ pCOHP analysis was performed using ONCV SG15 pseudopotentials^{51,52} using the Crawfish software package.⁵³ Further details are provided in Section V of the SI.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsmaterialslett.5c00857>.

Details of the machine learning model architecture and computational methods; an ablation study of the global- and node-level features; supplementary results on the electronic density of states and crystal orbital Hamilton population analysis (PDF)

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Notes

The authors declare no competing financial interest.

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